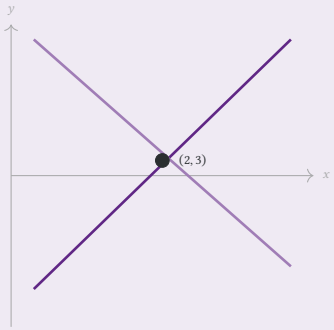
QR CODE **WATCH THE VIDEO BEFORE YOU SOLVE** YouTube • Lesson 2

RULE · Solution of a system – the crossing point

The solution is the (x, y) pair that makes **both** equations true – graphically, where the two lines cross.

Check: is $(2, 3)$ a solution?
$x + y = 5$: $2 + 3 = 5$ – true.
$2x - y = 1$: $2(2) - 3 = 1$ – true.
Both true – so yes, it's a solution.



RULE · Substitution method

If one equation already isolates a variable, plug that expression into the other equation.

$y = 2x - 1, \quad 3x + y = 9$ ← y is isolated

↓

$3x + (2x - 1) = 9$ ← substitute

↓

$5x - 1 = 9 \Rightarrow x = 2$ ← solve

↓

$y = 2(2) - 1 = 3 \Rightarrow (x, y) = (2, 3)$

RULE · Elimination method

Multiply one or both equations so matching coefficients cancel, then add or subtract.

$x + 2y = 11, \quad 3x + y = 18$ ← eliminate y

↓

$3x + y = 18 \ (\times 2) \Rightarrow 6x + 2y = 36$ ← multiply eq. 2 by 2

↓

$(6x + 2y) - (x + 2y) = 36 - 11 \Rightarrow 5x = 25$ ← subtract

↓

$x = 5 \Rightarrow y = 3 \Rightarrow (x, y) = (5, 3)$

RULES SHEET – 2/2

WARNING · Subtract EVERY term, not just one

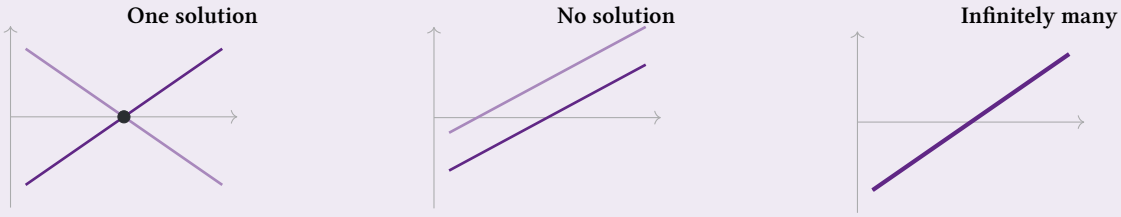
In $(6x + 2y) - (x + 2y)$, **every** term in the second parentheses changes sign – not just the first:

$6x + 2y - x - 2y = 5x + 0y = 5x$ **(dropping $2y$ fails to eliminate y!)**

RULE · Special cases: no solution & infinitely many

Compare the ratio of the x and y coefficients:

Ratios	Result
Coefficient ratios unequal	One solution
Coefficients equal, constants differ	No solution (parallel lines)
Coefficients and constants all equal	Infinitely many (same line)



Sign while solving: if eliminating a variable makes **both** variables vanish into a false statement ($0 = 12$) – **no solution**; if it's always true ($0 = 0$) – **infinitely many**.

WARNING · Always check what's being asked

$x + y = 10$, $x - y = 2$. **Question:** find xy. (Not x or y!)

Solve: $x = 6$, $y = 4$. The question asked for xy, so the last step is to multiply: $6 \times 4 = 24$.

Answer: $xy = 24$ – **not** $x = 6$ **or** $y = 4$.